

House Price Prediction in Bangalore Using a Neural Network

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Problem Statement

Goal:

Predict the price of a house in Bangalore using features like:

- Location
- BHK
- Total square feet
- No. of bathroooms

Why this problem?

- Real estate prices vary a lot in Bangalore
- Hard to estimate manually
- A model can learn patterns from existing data

Dataset & Preprocessing

Dataset Used

Bangalore House Price Dataset

Key Cleaning Steps

- Cleaned total_sqft (fixed ranges like 2100–2850, removed text like ‘15Acres’)
- Extracted BHK from “3 BHK”, “2 Bedroom”
- Removed rows with missing values
- Encoded location into numbers
- Scaled features (important for neural networks)

Features Used (Input To Model)

Final Features:

- Location
- BHK
- Total sq ft
- Bathrooms

Label (Output):

Final Price

Neural Network Model

Model Architecture:

- input Layer → 4 features
- Dense Layer (64 neurons, ReLU)
- Dense Layer (32 neurons, ReLU)
- Output Layer → 1 value (predicted price)

Training

- Optimizer: Adam
- Loss: Mean Squared Error (MSE)
- Epochs: 100

What the model does

How the model works

- Takes user inputs (location, BHK, sqft, bathrooms)
- Scales the values
- Passes them through the neural network
- Outputs an estimated price

Purpose

- Learn patterns from past house data
- Predict future house prices based on those patterns

Demo Output

Taking user inputs Location, BHK, Sq ft, Bathrooms

Conclusion:

- Neural networks can learn complex relationships
- Useful for price estimation in real estate
- Our model gives approximate price predictions based on key features

Libraries Used

- NumPy – Used for numerical calculations and handling arrays.
- Pandas – Used for loading, cleaning, and managing the dataset.
- Scikit-Learn – Used for train-test split, scaling features, and encoding locations.
- TensorFlow / Keras – Used to build and train the neural network model.
- Google Colab Files – Used to upload the dataset into Colab.

ReLU (Activation Function)

ReLU (Rectified Linear Unit) is an activation function used in neural networks. It keeps positive values as they are and turns negative values into zero. This helps the model learn faster and understand complex patterns more effectively.

The background of the image is a light-colored, marbled surface with intricate, swirling patterns in shades of grey, white, and a hint of pink. The texture is organic and fluid, resembling natural stone or a high-quality paper design.

Thank You!